Problem Set IV, Macroeconomics III
University of Copenhagen

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Exercise 1 – Comparative statics in the Diamond model

Consider the Diamond model from the lecture notes. Assume that utility is logarithmic and production is Cobb-Douglas, $f(k) = Ak^{\alpha}(0 < \alpha < 1)$. Describe how the following changes in parameters affect the capital accumulation schedule $k_{t+1}(k_t)$:

a) An increase in the rate of population growth, n.

b) A rise in A.

c) An increase in the share of capital in production, α.

Introduce depreciation in the Diamond model such that $r_t = f'(k_t) - \delta$ (δ is the rate at which capital depreciates).

d) For the case $\log$ utility, how does this affect the capital accumulation schedule $k_{t+1}(k_t)$?

Exercise 2 – An OLG model with intergenerational altruism

Consider the following OLG model: The preferences of generation t are

$$\mathcal{V}_t = \ln(c_{1,t}) + \beta \ln(c_{2,t+1}) + \xi \mathcal{V}_{t+1}$$

where ξ governs how much the current generation values the welfare of future generations. In particular,

$$\xi = \beta(1+n)\gamma$$

with $0 \leq \gamma < 1$ representing parental altruism. Further, assume that $\beta(1+n) < 1$. The budget constraints of agents born in period t are

$$c_{1,t} + s_t = w_t + b_t$$
$$c_{2,t+1} + (1+n)b_{t+1} = (1+r_{t+1})s_t$$

where $c_{1,t}$ and $c_{2,t+1}$ represent consumption in periods 1 (young) and 2 (old) of life, respectively; w_t is wage income of the young (the old do not work), s_t are savings which yield a return of r_{t+1}. When young, individuals receive bequest/inheritances from their parents (b_t) and when old, they may leave bequests themselves (b_{t+1}). Assume that b_{t+1} cannot be negative (you do not need to include this condition in your derivations).

(a) Through forward substitution, assuming that $\lim_{T\to\infty} \mathcal{V}_T < \infty$, show that

$$\mathcal{V}_t = \sum_{i=0}^{\infty} \xi^i \left(\ln(c_{1,t+i}) + \beta \ln(c_{2,t+1+i})\right)$$

(b) Set up the maximization problem for an agent born in period t. Solve for the first order conditions. Show that, in addition to the usual Euler equation, there is now an additional first order condition:

$$\frac{1}{c_{2,t+1}} = \gamma \frac{1}{c_{1,t+1}}$$

Comment on it's economic meaning. *(Hints: 1) remember that in comparison to the lecture notes, there is an additional choice variable now (b_{t+1}). 2) instead of setting up the Lagrangian, it may be more intuitive to substitute for c in the infinite sum – b_{t+1} influences the consumption decisions of future generations.*

(c) Find expressions for the steady state values of r and k. As altruism increases, what happens to interest rates and the capital stock? Discuss.

(d) If the government were to start spending wastefully, taxing each young generation a lump-sum of $\tau = g$, would this affect the capital stock? Why or why not?